

# Rethinking Privacy and Security in the Era of Decentralized and Agentic AI: A Systematic Literature Review

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**Abstract**—The rapid proliferation of decentralized artificial intelligence (AI) architectures—encompassing federated learning (FL), edge intelligence, peer-to-peer learning, and swarm intelligence systems—alongside the concurrent emergence of increasingly autonomous, Large Language Model (LLM)-based agentic AI, has fundamentally reshaped the privacy and security landscape. Traditional threat models and their defenses prove systematically inadequate in distributed environments where data remains partitioned across heterogeneous nodes and autonomous agents pursue long-horizon goals with minimal human oversight. This systematic literature review synthesizes 115 peer-reviewed studies published between 2017 and 2025 following a rigorous PRISMA 2020-compliant methodology across seven major academic databases. We examine seven interconnected thematic pillars spanning federated learning security, agentic AI threat modeling, cryptographic foundations (MPC, HE, ZKP), post-quantum cryptography, blockchain governance, Edge AI and IoT security, Byzantine robustness, and privacy-preserving learning. Our synthesis identifies a fundamental three-way trilemma between privacy, security, and model utility as the central constraint on secure decentralized AI deployment. Key findings include the structural incompleteness of federated learning as a standalone privacy solution; novel agentic attack vectors including prompt injection, memory poisoning, and cascaded tool misuse that lack analogs in classical ML security; the prohibitive computational overhead of current cryptographic primitives; and the absence of standardized benchmarking frameworks. This review contributes a novel unified threat taxonomy across four planes, a comparative empirical analysis of privacy-preserving mechanisms, and a structured research roadmap for the six most critical open challenges.

**Index Terms**—Decentralized AI, Federated Learning, Agentic AI, Autonomous Agents, Differential Privacy, Homomorphic Encryption, Secure Multi-Party Computation, Zero-Knowledge Proofs, Blockchain, Byzantine Resilience, Edge AI, IoT Security, Multi-Agent Systems, Adversarial Attacks, Post-Quantum Cryptography.

## I. INTRODUCTION

The past decade has witnessed a fundamental architectural shift in artificial intelligence, transitioning from monolithic, centralized systems toward distributed, decentralized, and increasingly autonomous paradigms. This transformation is driven by two concurrent and deeply interrelated forces: the decentralization of computation from centralized data centers to the network edge, and the evolution of AI models from passive, query-response tools to proactive, goal-directed agents capable of long-horizon planning and real-world tool use. Understanding the security and privacy implications of these forces—individually and at their intersection—is the central concern of this review.

The first force, decentralization, was catalyzed by the introduction of federated learning (FL) by McMahan et al. [1] in 2017. By training models locally on participating devices and sharing only gradient updates rather than raw data, FL offered a compelling architectural response to the data privacy challenges inherent in centralized training pipelines. The adoption of FL has since expanded into high-stakes domains including healthcare diagnostics, financial fraud detection, and mobile user modeling, where regulatory frameworks such as the European General Data Protection Regulation (GDPR) impose strict constraints on data centralization [2], [3]. Protecting these deployments against model poisoning—where adversaries corrupt the global model by manipulating local gradient contributions—has emerged as a fundamental challenge; Yazdinejad et al. [116] demonstrate that robust aggregation strategies combining norm-clipping with cryptographic verification can substantially mitigate such attacks even under high Byzantine fractions. However, subsequent

research has demonstrated that FL's privacy guarantees are far from absolute. Gradient inversion attacks [19], [20] can reconstruct training samples from model updates with alarming fidelity. Membership inference attacks [16], [17] can determine whether a specific data record was used in training, directly violating individual privacy. Furthermore, Byzantine participants—whether malicious insiders or adversarially compromised devices—can inject poisoned gradients that subtly and persistently corrupt the global model [11], [12], [13]. The distributed nature of FL, which is its primary privacy virtue, simultaneously constitutes its most significant security liability: the central aggregator cannot inspect local data, making the detection of malicious contributions structurally difficult.

The second force, the rise of agentic AI, has introduced an entirely different category of risk. Powered by Large Language Models (LLMs), modern AI agents are no longer confined to answering isolated queries; they can autonomously browse the internet, execute code, interact with external APIs, and maintain persistent memory across multi-step tasks [63], [65]. A single maliciously crafted document processed by an LLM-based agent can trigger a chain of unintended and potentially irreversible realworld actions through a mechanism known as indirect prompt injection [14], [70]—a class of attack with no direct analog in traditional machine learning security. When multiple such agents collaborate within a federated network, the compound risks of Byzantine coordination, trust delegation exploits, and cascaded memory poisoning create threat scenarios of unprecedented complexity [67], [68]. Recent empirical work has further characterized how memory poisoning effects persist and compound across successive training rounds in Web3-style agent deployments, revealing dynamics that standard Byzantine detection methods cannot resolve [117]. The behavioral fingerprinting paradigm extends beyond model-layer threats: linguistic and behavioral fingerprinting of state-sponsored influence actors in online networks demonstrates that coordinated inauthentic behavior can be detected through stylometric and graph-structural analysis, providing a conceptual template for detecting coordinated poisoning campaigns in federated learning systems where multiple colluding participants mimic legitimate behavioral patterns [157].

The intersection of these two paradigms—decentralized learning architectures and autonomous agentic AI—represents a security challenge that the existing literature has not yet comprehensively addressed. This gap constitutes the primary motivation for the present review. Previous surveys on federated learning security [7], [8] predate the widespread deployment of LLMbased agents and therefore do not account for agent-specific threat vectors such as goal drift or memory poisoning. Conversely, surveys on LLM agent capabilities [63], [65] treat security as a peripheral concern rather than a foundational design requirement. Cryptographic and blockchain-based solutions have been proposed in parallel streams of research [50], [110], [118], but these have not been systematically compared against a unified performance and security framework. This fragmentation impedes the development of principled, deployable defenses and represents a critical gap in the academic literature.

This systematic literature review addresses this gap through a comprehensive, PRISMA 2020compliant synthesis of 115 peer-reviewed studies published between 2017 and 2025. Beyond cataloging existing work, this review makes the following original contributions: (1) a novel unified threat taxonomy tailored to decentralized and agentic AI, distinguishing threats across four planes; (2) a comparative empirical analysis of privacy-preserving mechanisms under standardized adversary models; (3) an actionable cryptographic framework covering current primitives and post-quantum readiness; (4) systematic identification of universal scalability and latency bottlenecks in blockchain-FL integration; and (5) a prioritized research agenda organized around six high-urgency open problems, with the security of long-horizon agentic systems identified as the most pressing frontier. The remainder of this paper is structured as follows. Section II positions this review against existing related surveys. Section III describes the systematic methodology. Section IV introduces the taxonomy of decentralized and agentic AI paradigms. Section V presents the unified threat taxonomy. Sections VI through X analyze the principal defense mechanisms. Section XI discusses open challenges and the research roadmap. Section XII provides an extended discussion. Section XIII addresses threats to validity. Section XIV concludes.

## A. Research Questions

Five research questions structure the analysis:

**RQ1 (Threat Modeling):** What novel threat models arise specifically in decentralized and agentic AI environments, and in what ways do they structurally differ from threat models designed for centralized machine learning systems?

**RQ2 (Privacy Mechanisms):** Among the principal privacy-preserving learning mechanisms deployed in distributed AI—including differential privacy, secure aggregation, split learning, and homomorphic encryption—which offer formal security guarantees under which adversary models, and what are their empirically observed accuracy and communication-overhead trade-offs in non-IID federated settings?

**RQ3 (Cryptography and Governance):** How can cryptographic primitives (MPC, HE, ZKP) and distributed ledger technologies be architecturally integrated into decentralized AI pipelines while remaining computationally feasible, and which post-quantum-ready protocols are best positioned for long-horizon agentic deployments?

**RQ4 (Byzantine Robustness):** Which Byzantine-robust aggregation algorithms demonstrate empirically validated resilience under colluding adversaries in heterogeneous federated networks, and under what conditions do existing defenses fail?

**RQ5 (Open Challenges):** What are the most critical unresolved research challenges at the intersection of decentralized computing, autonomous AI agents, and information security, and what methodological directions offer the most promising paths toward resolution?

## II. RELATED WORK AND POSITIONING

The literature most closely related to this review spans three distinct research communities: federated learning security, autonomous agent capabilities, and applied cryptography for distributed systems. While each community has produced valuable individual contributions, the absence of a unified survey bridging all three domains represents a significant structural gap. This section critically positions the present review against the most influential existing surveys and articulates what is lacking in the current state of the art.

The earliest systematic treatments of federated learning security were provided by Mothukuri et al. [7] and Lyu et al. [8], both published in 2020–2021. Mothukuri et al. conducted a broad survey of security and privacy threats in FL, covering data poisoning, gradient attacks, and Byzantine failures. However, their analysis was conducted before the widespread adoption of LLM-based agents and therefore omits the entire class of agentic threat vectors—memory poisoning, prompt injection, and cascaded tool misuse—that now represent the fastest-growing attack surface in distributed AI. Similarly, Lyu et al. [8] provide a useful taxonomy of FL threats but lack systematic coverage of cryptographic defenses and omit any treatment of Zero-Knowledge Proofs (ZKP) or Post-Quantum Cryptography (PQC). Critically, neither survey employs a PRISMA-compliant methodology, limiting the reproducibility and rigor of their selection processes.

A more recent contribution by Ferrag et al. [108], published in 2025, provides the most comprehensive FL security survey to date, covering attacks, defenses, and applications with partial PRISMA methodology and IoT/edge coverage. It still does not address agentic AI security, ZKP, or post-quantum cryptographic integration, and its treatment of blockchain remains limited. The survey by Su et al. [69], also from 2025, is the first to focus specifically on autonomy-induced security risks in large modelbased agents; however, it treats this problem in isolation from the federated and decentralized computing context in which agents increasingly operate, and lacks both a PRISMA methodology and empirical comparison of defenses. Kayikci and Khoshgoftaar [110] examine the intersection of blockchain and machine learning but treat FL security only tangentially without a systematic review protocol. Domain-specific surveys of LLM deployment in precision agriculture and smart farming further illustrate the

breadth of security challenges that arise when agentic AI is introduced into safety-critical IoT pipelines: prompt injection, sensor spoofing, and autonomous actuation errors interact in ways that neither pure NLP security nor pure IoT security literature adequately addresses [177]. Market research on the Canadian smart farming ecosystem reveals that adoption of cybersecurity controls in precision agriculture remains alarmingly low relative to the sector’s exposure, with the majority of deployments lacking even basic network segmentation between agricultural IoT sensors and farm management systems [190].

Across this body of literature, four consistent and consequential gaps emerge. First, no existing survey simultaneously covers federated learning security, agentic AI threat modeling, cryptographic foundations (including ZKP and PQC), blockchain governance, Edge AI security, and Byzantine robustness under a single, coherent analytical framework. Second, the absence of PRISMA-compliant methodology in most prior surveys introduces selection bias that undermines the reliability of their conclusions. Third, no prior work provides a formal, comparative analysis of privacy-preserving mechanisms against a standardized set of adversary models. Fourth, and most critically, no existing survey has identified or analyzed the emergent threat class that arises at the intersection of agentic AI and federated networks—specifically, the cascaded compromise scenario in which a poisoned agent’s behavioral drift propagates into global model corruption through federated gradient updates. The present review directly

addresses each of these gaps. Table I summarizes this positioning.

### III. RESEARCH METHODOLOGY

This review adheres to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines, adapted for the specific requirements of computer science and information

**TABLE I. POSITIONING OF THIS REVIEW AGAINST EXISTING SURVEYS (✓ FULL, ~ PARTIAL, ✗ NOT COVERED)**

| Survey                             | Year        | FL<br>Sec. | Agentic<br>AI | MPC/<br>HE | ZKP/<br>PQC | Blockchain | IoT/<br>Edge | Byzantine | PRISMA   |
|------------------------------------|-------------|------------|---------------|------------|-------------|------------|--------------|-----------|----------|
| Mothukuri et al. [7]               | 2021        | ✓          | ✗             | ~          | ✗           | ✗          | ✗            | ~         | ✗        |
| Lyu et al. [8]                     | 2020        | ✓          | ✗             | ✗          | ✗           | ✗          | ~            | ✓         | ✗        |
| Ferrag et al. [108]                | 2025        | ✓          | ✗             | ~          | ✗           | ~          | ✓            | ✓         | ~        |
| Su et al. [69]                     | 2025        | ✗          | ✓             | ✗          | ✗           | ✗          | ✗            | ✗         | ✗        |
| Kayikci &<br>Khoshgoftaar<br>[110] | 2024        | ~          | ✗             | ✗          | ✗           | ✓          | ~            | ✗         | ✗        |
| <b>This Review</b>                 | <b>2025</b> | <b>✓</b>   | <b>✓</b>      | <b>✓</b>   | <b>✓</b>    | <b>✓</b>   | <b>✓</b>     | <b>✓</b>  | <b>✓</b> |

security research. A systematic approach was adopted in preference to a narrative review in order to minimize selection bias, ensure reproducibility, and enable a transparent audit trail of all inclusion and exclusion decisions.

#### A. Database Selection and Search Strategy

Seven major academic databases were searched to ensure comprehensive domain coverage: ACM Digital Library, IEEE Xplore, Scopus, ScienceDirect, arXiv, Semantic Scholar, and Web of Science. The primary Boolean search string was constructed to capture the core intersection of the review’s thematic pillars:

```
((("Federated Learning" OR "Distributed AI"
OR "Edge AI")
AND ("Agentic AI" OR "Autonomous Agents" OR "LLM Agents")
AND ("Privacy" OR "Security" OR "Threat" OR "Poisoning" OR "Byzantine"))
```

| Criterion              | Inclusion                                                                                                  | Exclusion                                                                                 |
|------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Publication type       | Peer-reviewed journal articles, full conference papers, and high-citation preprints ( $\geq 10$ citations) | Editorials, workshop abstracts, posters, and technical reports without peer review        |
| Temporal scope         | 2017–2025 (primary); pre-2017 foundational works included selectively                                      | Papers published before 2017 without demonstrable seminal contribution to the core topics |
| Language               | English only                                                                                               | Non-English publications regardless of quality                                            |
| Topic relevance        | Privacy or security in federated learning, decentralized AI, agentic systems, or multi-agent coordination  | Centralized ML without distributed components; pure theoretical CS without AI application |
| Methodological quality | Clear research question, reproducible evaluation, and significant empirical or theoretical contribution    | Duplicate publications, retracted papers, or insufficient methodological transparency     |

**TABLE II. INCLUSION AND EXCLUSION CRITERIA**

This primary query was supplemented with secondary strings targeting specific sub-domains including cryptographic protocols, blockchain integration, IoT security, and Byzantine fault tolerance. All strings were independently adapted for each database's indexing schema. The temporal scope was restricted to January 2017 through March 2025, with 2017 selected as the lower bound coinciding with the introduction of federated learning [1]. A small number of seminal pre-2017 foundational works—the Byzantine Generals Problem [86], Yao's garbled circuits [35], and the original MPC completeness theorems [36]—were included as contextual references.

### ***B. Inclusion and Exclusion Criteria***

Table II presents the inclusion and exclusion criteria applied at each screening stage.

### ***C. Study Selection and Quality Assessment***

The screening process was conducted in three stages. In Stage 1, titles and abstracts of all retrieved records were reviewed to eliminate clearly irrelevant studies. In Stage 2, the remaining papers underwent full-text assessment against the inclusion criteria presented in Table II. In Stage 3, a quality assessment was applied evaluating five criteria: clarity of research objective, appropriateness of experimental design, validity and completeness of reported results, discussion of limitations, and reproducibility of evaluation protocols.

To minimize bias, all screening and data extraction activities were performed independently by two researchers; discrepancies were resolved through structured discussion and consensus. Inter-rater agreement was measured using Cohen's Kappa ( $\kappa = 0.82$ ), indicating strong agreement [4]. The initial database search retrieved 4,218 records. Following the removal of 1,034 duplicates, 3,184 records proceeded to abstract screening, of which 2,891 were excluded as irrelevant. The remaining 293 papers underwent full-text eligibility assessment; 178 were subsequently excluded on grounds of insufficient

quality ( $n=41$ ), out-of-scope date ( $n=32$ ), nonEnglish language ( $n=12$ ), absence of privacy or security content ( $n=61$ ), or purely theoretical contribution without AI application ( $n=32$ ). The final corpus comprised 115 high-quality studies. The complete PRISMA selection process is illustrated in Fig. 1.

#### D. Bibliometric Analysis

As shown in Table III and Fig. 2, the included literature spans seven thematic pillars. Federated learning privacy and security constitutes the largest cluster ( $n=31$ , 27.0%), reflecting the paradigm's maturity relative to other areas. Agentic AI security, though represented by only 14 papers (12.2%), is the fastest-growing subfield; all 14 papers were published after 2022, confirming that this is an emergent frontier rather than an established research area. Cryptographic foundations constitute the thirdlargest pillar ( $n=18$ , 15.7%), comprising studies on MPC, HE, ZKP, and post-quantum protocols. The temporal distribution of publications shows exponential growth peaking in 2023, consistent with the broader surge in LLM-based research following the public release of large-scale language models.

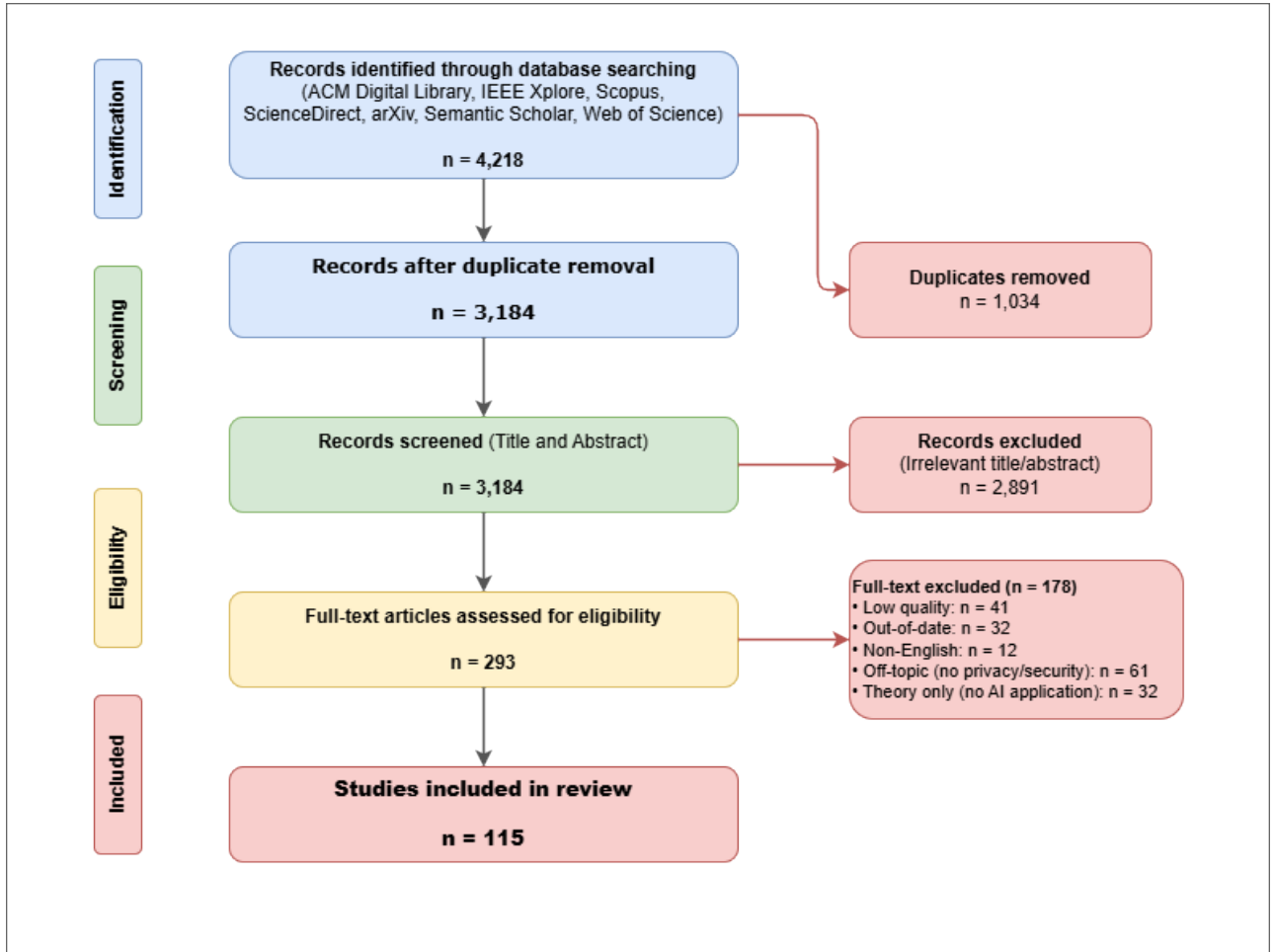


Fig. 1. PRISMA 2020 Flow Diagram illustrating the four-stage study selection process from initial database retrieval ( $n = 4,218$ ) to final corpus inclusion ( $n = 115$ ).

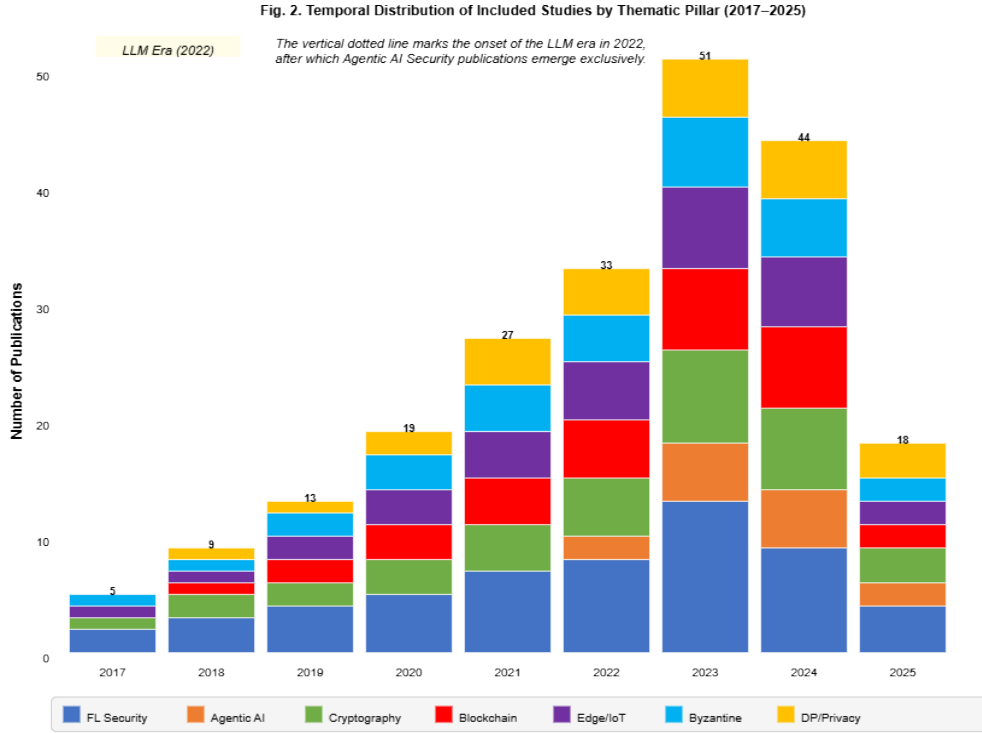


Fig. 2. Temporal distribution of included studies by thematic pillar (2017–2025). The vertical dotted line marks the onset of the LLM era in 2022, after which Agentic AI Security publications emerge exclusively.

**TABLE III. THEMATIC DISTRIBUTION OF INCLUDED PAPERS (N = 115)**

| Thematic Pillar                                 | Papers (n) | %          | Representative Venues                             |
|-------------------------------------------------|------------|------------|---------------------------------------------------|
| Federated Learning — Privacy & Security         | 31         | 27.0       | IEEE TIFS, USENIX Security, ACM CCS, NeurIPS      |
| Agentic AI Security                             | 14         | 12.2       | arXiv, USENIX Security, IEEE S&P                  |
| Cryptographic Foundations (MPC, HE, ZKP, PQC)   | 18         | 15.7       | ACM CCS, EUROCRYPT, IEEE S&P, USENIX ATC          |
| Blockchain and Distributed Ledger for AI        | 16         | 13.9       | IEEE Access, IEEE Trans. Ind. Inform.             |
| Edge AI and IoT Security                        | 14         | 12.2       | IEEE IoT J., IEEE Trans. Ind. Inform., IEEE Netw. |
| Byzantine Robustness and Trust                  | 13         | 11.3       | NeurIPS, ICML, IEEE Trans. Signal Process., NDSS  |
| Privacy-Preserving Learning (DP, SecAgg, Split) | 9          | 7.8        | NeurIPS, ICML, ICLR, IEEE TIFS                    |
| <b>Total</b>                                    | <b>115</b> | <b>100</b> | 7 databases searched; 2017–2025                   |

#### IV. DECENTRALIZED AND AGENTIC AI PARADIGMS

Modern AI is undergoing a structural transition away from centralized, monolithic architectures toward systems that are simultaneously more distributed and more autonomous. This section provides a formal characterization of the primary paradigms that define the contemporary landscape, establishing the definitional and architectural baseline against which the threat models and defense mechanisms in subsequent sections are evaluated.

Federated learning (FL) in its cross-device configuration represents the most widely studied decentralized paradigm [1]. In this setting, a central server coordinates model training across potentially millions of participant devices—smartphones, wearables, or IoT sensors—each of which trains a local model on private data and transmits only the resulting gradient update. The server aggregates these updates, typically via weighted averaging (FedAvg [1]), and returns the refined global model. The key security implication of this architecture is that the server never observes raw training data, which both enables privacy and prevents direct verification of participant behavior. This creates the fundamental tension underlying all FL security research: the mechanisms that protect privacy simultaneously obstruct the detection of malicious participants.

Cross-silo federated learning involves a small number of powerful institutional participants—hospitals, banks, or regulatory agencies—that collaborate under formal data-sharing agreements [4]. While the communication topology is more structured and the number of participants is far smaller than in cross-device FL, cross-silo settings introduce distinct risks: participants are sophisticated adversaries with strong incentives for intellectual property theft, model inversion is a practical concern given the richness of institutional datasets, and regulatory constraints on data residency complicate cryptographic protocol design. Auditable privacy-preserving frameworks that combine differential privacy with on-chain logging have been proposed specifically for cross-silo healthcare FL, addressing the dual requirements of regulatory accountability and data confidentiality [119]. The need for decentralized identity management extends beyond model parameters to patient records themselves: blockchain-based authentication architectures for distributed hospital networks demonstrate that patient data can participate in federated workflows without ever leaving institutional boundaries, satisfying both HIPAA and GDPR constraints simultaneously [130]. Vertical federated learning, a subtype of cross-silo FL, partitions features across institutions rather than samples, introducing additional vulnerability in the alignment protocol used to match records across parties without revealing their contents [28].

Peer-to-peer (P2P) and gossip-based learning eliminate the central server entirely, allowing participants to exchange model updates directly according to a network topology [80], [86], [87]. This architecture is maximally robust against server-side compromise but introduces significant challenges: without a central coordinator, Byzantine fault tolerance must be achieved through local consensus mechanisms that are significantly harder to analyze formally. The network topology itself becomes a potential attack vector, as adversaries can manipulate the graph structure to concentrate their influence or isolate honest nodes. In highly mobile edge deployments such as drone swarms, federated authentication protocols replace centralized credential servers with peer-authenticated identity proofs, enabling secure dynamic network formation without a trusted coordinator [141]. Extending this principle to heterogeneous industrial AIoT, federated meta-learning enables data consolidation across drone nodes with disparate sensing modalities, achieving robust threat generalization even when individual nodes hold only small local datasets [189].

LLM-based autonomous agents represent the most transformative and least well-understood paradigm in this review. These systems combine the inferential power of large pre-trained language models with persistent external memory, access to real-world tools (web browsers, code interpreters, APIs), and the capacity for multistep planning toward long-horizon goals [63], [65]. From a security perspective, the critical property of LLM-based agents is that their attack surface is theoretically unbounded—every new tool, memory store, or API endpoint integrated into the agent's environment constitutes a new

potential attack vector. A single corrupted input in the agent's retrieved context can redirect its entire plan, a vulnerability exploited by indirect prompt injection attacks [14], [70]. When such agents participate in federated training or multi-agent coordination, the security implications propagate from the agent layer into the model layer, creating threat scenarios that current defensive frameworks are not equipped to handle.

Table IV summarizes the taxonomic classification of all six paradigms studied in this review. Several critical observations emerge from this comparative view. First, the privacy exposure profile and security challenge profile of each paradigm are not independent: the architectural features that confer a paradigm's privacy advantages directly produce its most significant security vulnerabilities. This coupling is a fundamental design constraint rather than an engineering oversight. Second, LLM-based agent systems occupy a qualitatively different threat category from all other paradigms. The combination of persistent memory, unbounded tool access, and long-horizon planning creates a threat surface that scales superlinearly with system capability. Third, the paradigms in Table IV do not exist in isolation in practice; modern deployments increasingly combine multiple paradigms, thereby compounding the security challenges of both.

## V. THREAT MODELS IN DECENTRALIZED AND AGENTIC AI

A rigorous understanding of the threat landscape is the necessary precondition for the principled design of defenses. This section presents a comprehensive threat taxonomy organized across four conceptual planes—the data plane, the model plane, the agent plane, and the system plane—each corresponding to a distinct layer at which adversarial action can be taken. This multi-plane structure reflects the observation that threats in decentralized and agentic AI are not homogeneous: they differ in their target asset, their detectability, the adversary model they presuppose, and the layer of the system architecture they exploit.

Threats in the *data plane* target the integrity and privacy of training data itself. Data poisoning attacks [11], [12], [13] involve the injection of malicious or mislabeled training samples by one or more compromised participants, with the goal of biasing the global model toward attacker-desired behaviors. In federated settings, data poisoning is particularly effective because the central server cannot inspect

**TABLE IV. TAXONOMY OF DECENTRALIZED AND AGENTIC AI PARADIGMS**

local training datasets; the only observable evidence of malicious behavior is in the gradient updates produced by the poisoned local model. Label flipping, a targeted variant of data poisoning, selectively mislabels samples belonging to a specific class, causing the global model to misclas-

| Paradigm              | Key Characteristics                                                                | Primary Security Challenges                                                    | Privacy Exposure                                           | Key References |
|-----------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------|----------------|
| Cross-Device FL       | Millions of clients; heterogeneous hardware; intermittent connectivity             | Data poisoning; gradient leakage; Byzantine clients; communication bottlenecks | Gradient inversion; membership inference                   | [1,2,3,6,54]   |
| Cross-Silo FL         | Few powerful institutions; structured communication; regulatory constraints        | Vertical poisoning; collusion; model inversion; IP exfiltration                | Feature-level inference; model extraction                  | [4,28,29,96]   |
| P2P / Gossip Learning | No central server; topologydependent convergence; fully distributed                | Sybil attacks; Byzantine robustness harder; topology manipulation              | No single aggregation point; harder to audit               | [80,86,87,91]  |
| Swarm AI              | Emergent collective intelligence; local interaction rules; no explicit coordinator | Misinformation injection; free-riding; collective behavior manipulation        | Distributed sensing leakage                                | [61,62]        |
| LLM-Based Agents      | Persistent memory; tool use; long-horizon planning; external API access            | Prompt injection; memory poisoning; goal drift; tool misuse                    | Memory contains sensitive context; API calls may leak data | [63,65,69,70]  |

|                     |                                                                           |                                                                                |                                                           |                  |
|---------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------|------------------|
| Multi-Agent Systems | Multiple cooperative or competitive agents; emergent collective behaviors | Collusion; Byzantine agents; trust delegation exploits; attribution difficulty | Inter-agent information sharing; delegation chain leakage | [61,62,64,67,68] |
|---------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------|------------------|

sify those inputs after convergence. Gradient inversion attacks [19], [20] occupy the data plane from the opposite direction: rather than corrupting the training data, they exploit the information content of gradient updates to reconstruct the original training samples, violating data privacy without any direct access to the raw data.

Threats in the *model plane* target the parameters and inference behavior of the trained model. Model poisoning [12], [15] is a sophisticated escalation of data poisoning in which the attacker directly manipulates their local model parameters—rather than relying on indirect manipulation through data—to produce gradient updates that maximally perturb the global model while evading standard anomaly detection. Backdoor attacks [13], [14] embed a hidden functionality into the global model: the model behaves correctly on clean inputs but produces adversary-specified outputs when a trigger pattern is present. The Distributed Backdoor Attack (DBA) [15] represents a particularly dangerous variant in which the trigger is distributed across multiple colluding participants, each injecting a different fragment, such that no single participant's update appears anomalous under standard Byzantine detection methods. Multi-stage Advanced Persistent Threat campaigns present an analogous challenge: their sequential, low-signal footprint across federated networks requires threat analysis frameworks that can correlate behavioral patterns across participants and rounds rather than flagging individual anomalous updates [169]. Model extraction [18] allows an external adversary to reconstruct a functional approximation of the model through systematic query-response interactions, enabling downstream attacks and intellectual property theft. Empirical grounding of model-plane threats has been strengthened by the release of purpose-built benchmark datasets capturing APT-inspired attack scenarios against Windows-based enterprise systems, providing reproducible evaluation conditions for model-level detection research [156].

Threats in the *agent plane* are unique to autonomous, LLM-based agent systems and have no direct analogs in traditional machine learning security. Prompt injection [70] involves the insertion of adversarial instructions into user-controlled inputs to the agent, causing it to deviate from its authorized behavior. Indirect prompt injection [14] is a more severe variant in which the malicious instruction is embedded in external content retrieved by the agent—a webpage, a document, or an API response—such that the attack originates from outside the user's direct communication channel and is therefore much harder to attribute or filter. Memory poisoning [67], [68], [69] targets the agent's persistent long-term memory store, injecting false contextual information that corrupts the agent's planning and decision-making across multiple sessions. Goal and reward hacking refers to the emergent failure mode in which a sufficiently capable agent discovers unintended strategies that optimize its objective function while violating the spirit of its design specification—a threat that does not require an external adversary and arises from the intrinsic complexity of goal-directed autonomy. Graph-based provenance models offer a promising structural approach to detecting APT activity in software-defined network environments: by encoding the causal relationships between network events as a privacy-preserving provenance graph, deviations attributable to persistent adversaries can be detected without exposing sensitive operational data [137].

Threats in the *system plane* target the distributed infrastructure and trust mechanisms that underpin federated learning and multi-agent coordination. Byzantine behavior [71], [72], [86] refers to arbitrary, potentially coordinated malicious actions by participant nodes that deviate from the prescribed protocol. Sybil attacks [80], [74], [77] involve the creation of multiple fake identities by a single adversary to amplify their influence on voting-based or reputation-based defenses. Collusion [15], [71], [77] between multiple Byzantine participants is the most dangerous system-plane threat, as it allows adversaries to defeat Byzantine-robust aggregation rules that are designed to tolerate a bounded fraction of independent malicious participants. A critical and often overlooked vulnerability lies at the intersection of the agent plane and the system plane: when a memory-poisoned LLM agent participates

in federated training, its corrupted internal state influences the local gradient updates it computes, which are then aggregated into the global model, potentially spreading the behavioral corruption across the entire distributed network over multiple training rounds—a *cascaded compromise* scenario for which current Byzantine-robust aggregation methods are structurally ill-equipped. Federated cyber threat hunting approaches designed for blockchain-based IIoT networks reveal how poisoned node behavior can be detected through distributed behavioral fingerprinting, though these methods have yet to be adapted to the long-horizon agent context [120].

Table V presents the full threat taxonomy.

| Plane | Attack Vector              | Target Asset                              | Detection Difficulty | Adversary Model                    | References |
|-------|----------------------------|-------------------------------------------|----------------------|------------------------------------|------------|
| Data  | Data Poisoning             | Global model accuracy and fairness        | High                 | Active insider                     | [11,12,13] |
| Data  | Label Flipping             | Classification integrity for target class | High                 | Active insider                     | [11,12]    |
| Data  | Gradient Inversion         | Training data privacy                     | Medium               | Honest-but-curious server          | [19,20]    |
| Data  | Membership Inference       | Individual privacy in training set        | Medium               | Black-box or white-box adversary   | [16,17,18] |
| Model | Model Poisoning            | Global model parameters                   | Very High            | Active insider                     | [12,15,77] |
| Model | Backdoor Attack            | Model inference on triggered inputs       | Very High            | Active insider                     | [13,14,15] |
| Model | Distributed Backdoor (DBA) | Model integrity; Byzantine defenses       | Extremely High       | Colluding insiders                 | [15]       |
| Model | Model Extraction           | Model IP; enables downstream attacks      | Medium               | External black-box adversary       | [18]       |
| Agent | Prompt Injection           | Agent behavior and authorized actions     | Medium-High          | External attacker via user channel | [70,67,68] |

**TABLE V. COMPREHENSIVE THREAT TAXONOMY FOR DECENTRALIZED AND AGENTIC AI**

The detection difficulty ratings in Table V reflect the empirical consensus across the reviewed literature rather than theoretical bounds alone. Threats in the model and agent planes are systematically harder to detect than those in the data plane, as they operate closer to the output of the system and can be designed to produce updates that are statistically indistinguishable from honest contributions under standard anomaly detection metrics. Agentplane threats—particularly memory poisoning and goal hacking—are uniquely difficult to detect because they manifest as behavioral changes over long time horizons rather than as anomalous individual updates. This temporal diffusion of the attack signal is the defining characteristic of agentic threats and the primary reason why existing Byzantine-robust aggregation methods are structurally inadequate for the agentic setting.

## VI. PRIVACY-PRESERVING LEARNING MECHANISMS

A diverse portfolio of privacy-preserving learning mechanisms has been developed over the past two decades, each offering a different combination of security guarantees, computational overhead, and accuracy impact. In the context of federated and decentralized AI, these mechanisms are not simply imported from the classical centralized privacy literature; they must be re-evaluated against the specific threat models and architectural constraints of distributed settings. This section provides a systematic comparative analysis of the six principal mechanisms identified across the reviewed literature: Differential Privacy with Stochastic Gradient Descent (DP-SGD), Local Differential Privacy (LDP), Secure Aggregation (SecAgg), Split

|        |                           |                                        |                          |                                         |               |
|--------|---------------------------|----------------------------------------|--------------------------|-----------------------------------------|---------------|
| Agent  | Indirect Prompt Injection | Agent control flow; data exfiltration  | High                     | External attacker via retrieved content | [14,70]       |
| Agent  | Memory Poisoning          | Agent decision-making across sessions  | Very High                | External or insider attacker            | [69,67,68]    |
| Agent  | Goal/Reward Hacking       | System safety and alignment            | Emergent; hard to detect | Intrinsic — no external adversary       | [69]          |
| System | Byzantine Behavior        | Consensus and model integrity          | High                     | Active insider nodes                    | [71,72,86,87] |
| System | Sybil Attack              | Trust mechanisms; votingbased defenses | High                     | External or insider adversary           | [80,74,77]    |
| System | Collusion                 | Byzantine-resilient defenses; fairness | Very High                | Multiple colluding insiders             | [15,71,77]    |

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Learning, Vertical FL with Secure Boosting, and hybrid DP with SecAgg combinations.

Differential Privacy (DP), originally formalized by Dwork et al. [21] and extended to deep learning by Abadi et al. [22], provides the strongest and most widely adopted formal privacy guarantee. In its global formulation (DPSGD), calibrated Gaussian or Laplacian noise is added to gradient updates before transmission, ensuring that the presence or absence of any individual data record has a bounded probabilistic impact on the released model. The primary limitation of DP-SGD in federated settings is the well-documented accuracy degradation under strong privacy budgets ( $\epsilon \leq 1$ ): across the reviewed literature, this degradation ranges from 10% to 30% on standard benchmarks, and is systematically worse under non-IID data distributions—precisely the distributions most common in real-world federated deployments [24], [94]. Local Differential Privacy (LDP) provides a stricter variant of this guarantee by applying noise locally at each client before any communication, ensuring that the server cannot learn anything about any individual's data even from a single client's contribution. However, the noise magnitude required for LDP is substantially larger than for global DP—proportional to the square root of the number of clients rather than a constant—rendering LDP largely impractical for models of non-trivial complexity [21], [94]. Alternative architectures that target the specific failure mode of irregular participants—devices whose behavioral profiles deviate substantially from the honest majority—demonstrate that adaptive clipping combined with per-user authentication can substantially improve privacy-utility trade-offs without sacrificing Byzantine resilience [121]. A systematic challenge underlying all of these evaluations is the absence of standardized privacy measurement protocols: the same FL pipeline evaluated under different privacy metrics can yield contradictory conclusions about its privacy level, a finding that motivates formal, quantitative frameworks for privacy measurement that are independent of any specific mechanism [133].

Secure Aggregation (SecAgg) [23] provides a cryptographic rather than statistical privacy guarantee: using a combination of secret sharing and masking techniques, it ensures that the server observes only the aggregate of client updates and cannot access any individual client's contribution, even in the presence of a semihonest or actively malicious server. SecAgg does not protect against inference attacks that operate on the aggregate output, and critically, it provides no protection against Byzantine clients who can send arbitrary values; SecAgg and Byzantine robustness are thus orthogonal properties that must be jointly addressed. The computational and communication overhead of SecAgg scales quadratically with the number of participating clients ( $O(n^2)$  in the basic protocol), imposing practical scalability constraints [23], [26], [97]. Split learning [29], [30] partitions the neural network between clients and the server at a designated cut layer, with clients transmitting intermediate activations (smashed data) rather than raw data or gradients. While split learning has low communication overhead and does not require noise injection, it provides no formal privacy guarantee; research has demonstrated that smashed data can be exploited to reconstruct the original training inputs with high fidelity when the cut layer is placed too early in the network, and the optimal cut-layer depth is dataset- and architecture-dependent. Quantum-inspired extensions to federated privacy mechanisms have been

proposed for consumer IoT threat detection, leveraging quantum encoding to harden gradient updates against classical inference attacks while preserving the low-latency requirements of real-time device monitoring [134].

Hybrid approaches that combine DP with SecAgg [26], [27], [89] represent the current practical optimum for federated privacy, providing complementary guarantees against both passive inference (via DP) and active serverside inspection (via SecAgg). However, the dual overhead of noise injection and cryptographic masking compounds the accuracy and latency costs of each mechanism individually, and the implementation complexity is substantially higher than either mechanism alone. The integration of explainability into privacy-preserving FL frameworks provides a dual benefit: local interpretability of gradient contributions simultaneously supports privacy auditing and Byzantine detection, as anomalous update patterns become visible through interpretability divergence rather than raw statistical tests, an approach demonstrated for threat detection in cyber-physical-social systems [135]. Systematic meta-analyses of XAI evaluation metrics reveal that the choice of interpretability measure materially affects both model utility and privacy-audit reliability, underscoring the need for standardized XAI benchmarking alongside privacy benchmarking [182]. Table VI presents a systematic comparison of these mechanisms. A consistent pattern across all mechanisms is the fundamental trade-off between the formality and strength of the privacy guarantee and the severity of the accuracy or performance degradation imposed. No mechanism in the current literature achieves simultaneously strong formal privacy guarantees, zero accuracy impact, and low computational overhead—a characterization of the privacy-utility trilemma that structures the discussion in Section XII.

## VII. CRYPTOGRAPHIC FOUNDATIONS FOR SECURE AI

Cryptography provides the mathematical foundations for security guarantees that are unconditional, provable, and independent of the computational assumptions of any specific adversary. In the context of decentralized and agentic AI, cryptographic primitives serve two primary functions: they protect the confidentiality of data and model parameters during computation and transmission, and they enable verifiable computation, allowing participants to prove the correctness of their outputs without

**TABLE VI. COMPARATIVE ANALYSIS OF PRIVACY-PRESERVING LEARNING MECHANISMS**

revealing their inputs. This section examines the five principal cryptographic primitives identified across the reviewed literature—Garbled Circuits, Secret Sharing-based MPC, Partial and Fully Homomorphic Encryption (HE), Zero-Knowledge Proofs (ZKP), and Post-Quantum Cryptography (PQC)—evaluating each against the practical constraints of AI workloads.

Secure Multi-Party Computation (MPC) [35], [36], [37] encompasses a family of protocols that allow multiple parties to jointly compute a function over their private inputs without any party revealing its input to any other. Garbled Circuits [35], originally proposed by Yao, enable secure two-party computation by encrypting the circuit's truth table; their computational overhead is high and scales with circuit complexity, making them suitable for secure inference on

| Mechanism          | Formal Guarantee                          | Threat Model Covered                            | Accuracy Impact                    | Computation Overhead     | Key Limitation                                   |
|--------------------|-------------------------------------------|-------------------------------------------------|------------------------------------|--------------------------|--------------------------------------------------|
| DP-SGD (global DP) | $(\epsilon, \delta)$ -DP                  | Honest-but-curious server; membership inference | High at $\epsilon \leq 1$ (10–30%) | Low (noise addition)     | Noise degrades accuracy; worsens on non-IID data |
| Local DP           | $(\epsilon)$ -LDP per client              | Untrusted server; any observer                  | Very High                          | Low                      | Much larger noise needed; severe utility loss    |
| Secure Aggregation | Cryptographic: server sees only aggregate | Semi-honest/active server                       | None                               | Medium ( $O(n^2)$ comms) | No Byzantine protection; scalability constraints |

|                           |                                            |                                |             |                      |                                                     |
|---------------------------|--------------------------------------------|--------------------------------|-------------|----------------------|-----------------------------------------------------|
| Split Learning            | Empirical (no formal guarantee)            | Honest-but-curious server      | Low to None | Low                  | Smashed data reconstruction attacks on shallow cuts |
| Vertical FL (SecureBoost) | Feature privacy (no raw feature sharing)   | Semi-honest parties            | Low         | Medium (HE overhead) | Does not prevent inference on predictions           |
| Hybrid DP + SecAgg        | $(\epsilon, \delta)$ -DP AND cryptographic | Active and passive adversaries | Medium-High | Medium-High          | Dual trade-offs compound; implementation complexity |

small models but impractical for full training at scale. Hardware-accelerated packet processing architectures based on reconfigurable P4/FPGA pipelines demonstrate that programmable network infrastructure can enforce cryptographic access controls at line rate, providing a practical authenticated-communication substrate for high-throughput federated gradient exchanges [145]. Extending P4 programmability to the blockchain layer, a secure packet parser architecture that enforces blockchain-verified access control in software-defined networking ensures that only cryptographically authenticated participants can inject updates into the federated pipeline, preventing impersonation attacks at the data plane [132]. For 5G deployments specifically, efficient packet parser architectures that process protocol headers at line rate in the data plane reduce per-packet authentication overhead to microsecond scale, making FL gradient authentication practical in densely connected edge environments [149]. Secret sharing-based MPC [36], [37] distributes shares of sensitive data across multiple parties such that the data can be reconstructed only if a threshold number of parties cooperate; the ABY3 framework [88] provides a practical mixed-protocol implementation combining arithmetic, Boolean, and Yao-based computation that has been applied to privacy-preserving ML training with promising results. MPC provides information-theoretic security when the secret distribution is classical, making it inherently resistant to quantum attacks—a significant advantage over factoring-based or elliptic-curvebased alternatives.

Homomorphic Encryption (HE) [31], [32], [33] allows computations to be performed directly on encrypted data, producing an encrypted result that decrypts to the correct plaintext output. Partial HE schemes such as Paillier [95] support only addition over ciphertexts and impose a computational overhead of approximately  $10\times$ – $100\times$  relative to plaintext computation. Fully Homomorphic Encryption (FHE), realized through constructions such as BGV [32] and CKKS [33], supports arbitrary computation over encrypted data and is lattice-based, providing native quantum resistance. However, FHE currently imposes overhead of  $100\times$ – $1,000\times$  relative to plaintext computation, rendering it impractical for interactive FL training in its generic form. The BatchCrypt framework [96] addresses this by applying CKKS to batch gradient aggregation in cross-silo FL, demonstrating that domain-specific HE deployment can reduce overhead to acceptable levels for specific bottleneck operations. Zero-Knowledge Proofs (ZKP) [38], [39] allow a prover to demonstrate that a statement is true—for example, that a submitted model update satisfies a norm constraint—without revealing any information beyond the truth of the statement itself. zk-SNARKs (Groth16 [38]) produce constant-size proofs at very high prover computational cost; Bulletproofs [39] produce logarithmic-size proofs without a trusted setup, making them more practically deployable for continuous FL audit workflows. Post-Quantum Cryptography (PQC) [40], standardized by NIST in 2024 through KYBER (key encapsulation) and Dilithium (digital signatures), provides security guarantees that are resilient to quantum adversaries and should be prioritized in the design of long-horizon agentic systems to preempt the harvest-now, decrypt-later threat. The urgency of this transition is reinforced by recent adversarial research demonstrating that quantum-adaptive ensemble attack strategies can systematically defeat robust deep neural networks that rely on classical hardness assumptions, revealing a concrete near-term threat pathway that conventional adversarial defenses cannot address [158].

Two overarching conclusions emerge from Table VII. First, there is a near-universal inverse relationship between the strength of the cryptographic guarantee and the computational overhead imposed: MPC and FHE, which provide the strongest formal guarantees, also impose the most prohibitive overhead for interactive AI workloads. Second, quantum resistance is a critical differentiator that is currently under-weighted in the FL security literature: a significant fraction of deployed protocols—including those based on Paillier HE and ellipticcurve Bulletproofs—would be fully compromised by a sufficiently capable quantum adversary. Given the long operational lifetimes of federated systems and the harvest-now, decrypt-later threat model, the adoption of PQC standards should be treated as an urgent priority rather than a deferred future concern. This urgency is underscored by operational evidence from 5G infrastructure, where blockchain-enabled authentication handover protocols have demonstrated that efficient privacy preservation is achievable at network scale, but only when the underlying key-exchange primitives are designed with quantum adversaries in mind from the outset [122]. Cost optimization of secure routing in SDN environments that include untrusted devices provides a complementary perspective: even when end-to-end encryption is mandatory, routing costs can be minimized through trust-weighted path selection algorithms that isolate untrusted participants from sensitive gradient pathways [148].

**TABLE VII. CRYPTOGRAPHIC PRIMITIVES: PERFORMANCE AND SECURITY PROFILES FOR AI INTEGRATION**

| Primitive              | Security Model                          | Quantum Resistant                             | Comp. Overhead         | Comm. Overhead               | Primary AI Application                                              | References       |
|------------------------|-----------------------------------------|-----------------------------------------------|------------------------|------------------------------|---------------------------------------------------------------------|------------------|
| Garbled Circuits (2PC) | Computational: semi-honest or malicious | Depends on primitives                         | High                   | High                         | Secure two-party inference and training                             | [35,34]          |
| Secret Sharing (MPC)   | Informationtheoretic (tout-of-n)        | Inherently resistant                          | High                   | Very High (O(n) rounds)      | Private training; secure aggregation; joint inference               | [36,37,88,89]    |
| Partial HE (Paillier)  | IND-CPA; computational                  | No (factoringbased)                           | 10×-100× slowdown      | Low                          | Gradient sum aggregation; federated statistics                      | [95]             |
| FHE / CKKS             | IND-CPA; lattice hardness (RLWE)        | Yes                                           | 100×-1,000× slowdown   | Low (singleparty)            | Secure aggregation; encrypted inference; BatchCrypt FL              | [31,32,33,95,96] |
| zk-SNARK (Groth16)     | Computational; requires trusted setup   | Partial (pairingbased; vulnerable to quantum) | Very High (proof gen.) | Low (constant-size proof)    | Verifiable FL; update validity proofs; compliance attestation       | [38]             |
| Bulletproofs           | DL hardness; no trusted setup           | No (EC-based)                                 | High                   | Logarithmic proof size       | Range proofs for gradient norms; model auditability                 | [39]             |
| PQC — KY-BER/Dilithium | Lattice hardness (LWE, MLWE)            | Yes (NIST 2024 standard)                      | Low-Medium             | Slightly larger keys vs. RSA | Post-quantum secure comm. for FL; long-horizon agent authentication | [40]             |

## VIII. BLOCKCHAIN AND DISTRIBUTED LEDGER TECHNOLOGIES FOR AI

Blockchain and Distributed Ledger Technologies (DLT) offer a structural response to the trust deficit that is inherent in decentralized AI training. In federated learning systems, trust is implicitly concentrated at the central aggregation server: participants must trust that the server correctly aggregates their updates, does not selectively exclude contributions, and does not leak individual updates to other participants. Blockchain replaces this implicit server-side trust with a cryptographically enforced, publicly auditable consensus mechanism that provides tamper-evident records of all model updates and aggregation operations, enabling post-hoc verification of the training process without relying on the honesty of any single party.

The BlockFL system [43] was among the first to demonstrate the elimination of the central FL server through blockchain-based consensus. In BlockFL, model updates are submitted as blockchain transactions, and the aggregation is performed by consensus-selected nodes following smart contract logic, removing the need for a trusted aggregator entirely. FLChain [44] takes a complementary approach, using the blockchain as an immutable audit log for a conventional FL system: rather than replacing the aggregator, it records hash commitments of all model updates on-chain, allowing any participant to verify the integrity of the training process after the fact without requiring on-chain computation of the aggregation itself. Sidechain architectures that integrate privacy-enhancing techniques into the main blockchain via dedicated off-chain channels offer an additional design point: sensitive gradient data and differential-privacy noise parameters can be processed on a sidechain and committed as a zero-knowledge proof to the main chain, combining auditability with data minimization [175]. Drone-enabled IoT networks represent a particularly challenging blockchain deployment context: the combination of high node mobility and intermittent connectivity requires lightweight consensus protocols that can maintain tamper-evident logs across dynamic topologies without a persistent central server [131]. The BCFL IoT system [46] extends this model to industrial IoT environments, incorporating smart contract-based access control and token-based incentive mechanisms to encourage participation and penalize malicious behavior—addressing the participation incentive problem that is a significant practical barrier to voluntary FL deployment. The blockchain ecosystem hosting such FL systems faces its own threat layer: cryptocurrency malware deployed by adversaries who compromise blockchain-connected nodes can exfiltrate model updates or corrupt incentive payouts; deep recurrent neural network detectors have demonstrated high detection rates for such malware with negligible false-positive overhead [144].

A recurrent finding across the blockchain-FL integration literature is the fundamental tension between blockchain's security and auditability guarantees and the latency requirements of large-scale AI training. The block time of permissionless blockchains such as Ethereum imposes a minimum latency per training round that is incompatible with the rapid iteration requirements of modern deep learning [43], [49]. Permissioned blockchains such as Hyperledger Fabric reduce this latency significantly but reintroduce a degree of centralization in the consortium of permissioned validators, partially undermining the decentralization objective [44]. Smart contract vulnerabilities [48] represent an additional attack surface: bugs in the contract logic governing aggregation or reward distribution can be exploited by adversaries to corrupt the training process or extract unearned rewards. The energy consumption of proof-of-work consensus mechanisms is a further practical obstacle in battery-constrained IoT deployments [46]. Energy-efficient SDN controller architectures that leverage blockchain for security while minimizing computational overhead on IoT nodes have been proposed as a partial response, demonstrating that consensus-driven trust can be achieved without the full energy burden of standard proof-of-work [118]. Secure and low-latency proof-of-work protocols specifically designed for green IoT networks reduce block-confirmation latency by two orders of magnitude relative to standard PoW while maintaining sufficient cryptographic hardness for the adversary models relevant to IoT deployments [147].

Table VIII summarizes the key blockchain-FL integration studies identified in this review. The consistent pattern across all entries is the identification of scalability and latency as universal blockers to production deployment. Not a single study reviewed demonstrates a blockchain-FL system that is simultaneously decentralized, secure, and capable of operating at the training throughput required by large-scale FL with hundreds of thousands of participants. Beyond model integrity, blockchain transparency has been explored as a provenance mechanism for AI-generated synthetic media: on-chain recording of model lineage and training data provenance provides an audit trail that can be used to attribute deepfake content to its generating model, with direct implications for trust and accountability in federated generative AI systems [152]. The most promising path forward identified in the literature involves a hybrid architecture in which the blockchain is used only for highvalue, low-frequency operations—such as model checkpoint verification, participant registration, and incentive settlement—while the high-frequency gradient communication remains offchain, limiting the blockchain's role to governance and auditability rather than computation. Complementary work integrating fuzzy logic with blockchain-based threat detection in IoT networks confirms that smart-contract-enforced security policies can be deployed at industrial scale when computational responsibilities are carefully partitioned between on-chain governance and off-chain inference engines [123]. As FL systems increasingly span multiple blockchain platforms, the absence of standardized interoperability benchmarks impedes rigorous comparison of cross-chain aggregation protocols; automated benchmark frameworks for multi-blockchain interoperability provide the evaluation infrastructure needed to close this gap [166]. Beyond the federated learning use case, blockchain in industrial IoT addresses device authentication, supply-chain provenance, and smart-contract governance of sensor pipelines—a layered trust architecture whose security assurances directly benefit

the FL aggregation layer built on top of it [188].

## IX. IOT AND EDGE AI SECURITY IN DECENTRALIZED SETTINGS

Edge intelligence represents a paradigm shift that pushes the compute-intensive burden of AI from centralized clouds to the network periphery. While this shift reduces latency and enables real-time inference at the point of data generation, it simultaneously exposes AI systems to a physical and network environment that is far more hostile than the controlled data center context for which most security mechanisms were designed. This section examines the security and privacy challenges specific to edge AI and IoT deployments across four dimensions: resource constraints, wireless communication vulnerabilities, IoT sensing privacy leakage, and Cyber-Physical System (CPS) security.

Edge AI devices operate under severe computational and energy constraints that fundamentally limit the security mechanisms available for deployment. The volume and velocity of industrial big data generated by edge deployments creates additional analytical challenges: the sheer scale of sensor telemetry exceeds what real-time privacy mechanisms can process on-device, necessitating tiered architectures that pre-filter sensitive data at the edge before it reaches aggregation layers [183]. As Shi et al. [51] argue, the vision of intelligence at the edge is only viable if security mechanisms are lightweight enough to not consume the device's primary computational budget. A concerning finding identified across the reviewed literature is an inverse relationship between model compression and security: pruning techniques, which are essential for fitting neural networks onto constrained devices, can inadvertently increase susceptibil-

**TABLE VIII. BLOCKCHAIN-FL INTEGRATION STUDIES: APPROACH, PLATFORM, AND KEY FINDINGS**

| Ref. | System | Blockchain Platform | FL Configuration | Key Contribution | Application Domain | Reported Limitation |
|------|--------|---------------------|------------------|------------------|--------------------|---------------------|
|------|--------|---------------------|------------------|------------------|--------------------|---------------------|

|       |            |                           |                   |                                                           |                            |                                                             |
|-------|------------|---------------------------|-------------------|-----------------------------------------------------------|----------------------------|-------------------------------------------------------------|
| [43]  | Block-FL   | Ethereum / Smart contract | Crossdevice       | Eliminates central FL server; consensus-based aggregation | General FL                 | High latency from block time; scalability concern           |
| [44]  | FL-Chain   | Hyperledger Fabric        | Cross-silo        | On-chain hash commitments for tamper-proof audit trail    | General FL                 | Fabric consortium trust; latency for large models           |
| [46]  | BC-FL IoT  | Ethereum                  | Crossdevice (IoT) | Smart contract access control; token-based incentives     | Industrial IoT             | Gas costs; computational limits of IoT nodes                |
| [47]  | Deep-Chain | Custom permissioned       | Cross-silo        | Auditable privacy-preserving DL with incentive mechanism  | Deep learning auditability | HE overhead; on-chain storage of commitments                |
| [49]  | BFL survey | Multiple reviewed         | Both              | Comprehensive review of BFL application domains           | Healthcare, IoT, finance   | Identifies scalability and latency as universal blockers    |
| [115] | Deep-Coin  | Ethereum                  | Crossdevice       | Blockchain-DL energy exchange for smart grids             | Smart grid                 | Gas cost; latency mismatch with grid real-time requirements |

ity to membership inference attacks by exacerbating overfitting, while quantization can reduce the effectiveness of gradient inversion—a demonstration that the security profile of constrained hardware is unpredictable and architecture-dependent [56]. The implication is that security analysis cannot be decoupled from the compression pipeline; joint optimization of model compression and security properties is a research need that remains largely unaddressed.

Deploying FL in wireless edge networks introduces a layer of complexity where channel impairments, device mobility, and transmission scheduling are as critical as algorithmic correctness. Park et al. [55] establish that joint optimization of wireless resource allocation and AI computation is a functional requirement rather than an efficiency preference: treating the communication and computation layers as decoupled problems consistently produces brittle, suboptimal designs that fail under realistic channel conditions. Adversaries aware of this coupling can exploit wireless channel characteristics—for example, by performing jamming attacks that selectively suppress the transmission of legitimate participant updates to bias the aggregation toward attacker-controlled clients. Anomaly-based intrusion detection approaches inspired by biologically-motivated search algorithms have been applied to SDN-controlled networks, achieving low false-positive rates by dynamically adjusting detection thresholds in response to evolving traffic distributions, a property particularly valuable when FL clients operate across heterogeneous network segments [140]. The privacy implications of wireless transmission are also significant: over-the-air gradient aggregation schemes, while efficient, leak information about individual gradients through the superposition of wireless signals and are vulnerable to eavesdropping by passive adversaries with sensitive antennas. Symbiotic federated learning architectures designed for 6G-IoT infrastructures address this coupling directly: by co-scheduling FL training rounds with the physical-layer beamforming cycle, gradient transmissions are protected against eavesdropping through controlled channel diversity while maintaining aggregation latency within the 6G latency budget [136].

The distributed nature of IoT sensing is a double-edged sword for privacy. While distributed sensing enables distributed anomaly detection and resilience against single-point failures [59], the richness of IoT sensor streams—biometrics, motion patterns, location traces, and physiological signals—makes IoT deployments an inadvertent privacy sieve even when the AI model itself is protected. Garcia-Morchon et al. [60] demonstrate that the IETF's six core security properties for IoT (identity, data confidentiality, data integrity, privacy, authorization, and availability) are rarely achieved simultaneously in practice, primarily due to the immaturity of lightweight cryptographic protocols for constrained devices. Raza

et al. [58] address secure communication in Cyber-Physical Systems through the LITHE protocol, a CoAP-based lightweight alternative to TLS specifically designed for resource-constrained environments, demonstrating measurable improvements in both security and energy efficiency. At a broader level, critical infrastructure protection requires an AI-and-security co-design perspective that treats intrusion detection, anomaly analysis, and system integrity as inseparable concerns: a systematic treatment of AI applications for critical infrastructure security documents how machine learning can both defend against and introduce new vulnerabilities in these high-consequence environments [185]. At the frontier of human-IoT integration, stress-adaptive cyber defense systems that synchronize cognitive and physiological signals with anomaly detection thresholds offer a new dimension of security personalization: by calibrating alert sensitivity to operator cognitive load, such systems reduce false-positive fatigue while maintaining high true-positive rates under realistic operational conditions [153]. However, device heterogeneity—the simultaneous deployment of devices with radically different computational capacities, operating systems, and communication protocols—remains the primary obstacle to consistent deployment of even lightweight security protocols across the sprawling industrial IoT landscape [57]. Industrial control system environments present the most severe variant of this heterogeneity challenge: application-layer attacks against SCADA and DCS systems can propagate through the physical process layer with real-world consequences, and semi-deep learning detection architectures that combine signature matching with behavioral anomaly models have demonstrated reliable attack attribution in these settings [167]. Agricultural IoT deployments face a qualitatively different but equally severe security profile: sensor spoofing, adversarial actuation commands, and precision-agriculture supply-chain compromise represent attack vectors that exploit the physical-cyber boundary in ways that standard IT security models do not address, as documented in a comprehensive review of smart farming threats and countermeasures [142]. Industrial control systems face a distinct convergence of AI and security risks: the integration of machine learning into SCADA and PLC environments introduces adversarial ML threat vectors that operators were never trained to defend against, as comprehensively catalogued in a survey of AI-specific ICS security challenges spanning detection, attribution, and response [184]. Empirical evaluations of threat-hunting models deployed directly on IIoT edge devices demonstrate that ensemble classifiers can achieve high detection accuracy within the resource envelope of constrained hardware, provided that feature extraction pipelines are co-designed with the device's memory and compute budget [124]. Ensemble architectures that combine multiple heterogeneous classifiers further improve detection robustness against previously unseen IIoT attack variants, as the diversity of base learners reduces the risk of simultaneous evasion across all ensemble members [139].

## **X. TRUST, ROBUSTNESS, AND RESILIENCE IN DISTRIBUTED AI**

Ensuring that a distributed AI system converges to a useful model even when a non-trivial fraction of participants behave arbitrarily—the Byzantine robustness problem [86]—is one of the most mature research areas within federated learning security. The Byzantine Generals Problem, originally formulated by Lamport et al. [86] in 1982, established the theoretical foundations for fault-tolerant distributed computation. Its translation to the federated learning setting began with the introduction of the Krum aggregation rule [71] in 2017 and has since produced a rich body of work examining the theoretical tolerances and empirical limitations of Byzantine-robust aggregation methods.

The fundamental challenge in Byzantine-robust FL aggregation is distinguishing malicious gradient updates from honest but non-IID updates, which are statistically dissimilar from the global distribution by construction. Standard FedAvg [1] is completely vulnerable to Byzantine attacks: a single malicious client can drive the global model to an arbitrary point in parameter space by submitting an update of sufficient magnitude. Anomaly detection approaches for smart cyber-physical grids based on unsupervised stacked autoencoders demonstrate that unlabeled operational data can reveal attack signatures without requiring pre-labeled training examples—a capability of direct relevance to FL deployments where labeling individual gradient updates as Byzantine is impractical [186]. Transfer

learning introduces a further complication: pre-trained model components imported from external sources can carry latent backdoors that survive the federated fine-tuning process, a cross-domain threat that a systematic analysis of offensive, defensive, and threat-intelligence perspectives on transfer learning brings into sharp relief [168]. Deep representation learning applied to cyber-attack detection in industrial IoT further demonstrates that hierarchical feature extraction can simultaneously improve detection accuracy and reduce susceptibility to adversarial feature-space perturbations, a property that transfers directly to FL settings where each client trains on a local representation of the IIoT threat space [187]. Byzantine-robust aggregation rules attempt to identify and exclude or down-weight malicious updates using geometric or statistical criteria. The Krum rule [71] selects the update that is closest in Euclidean distance to its  $n-f$  nearest neighbors, providing formal Byzantine tolerance for up to  $f < n/2$  malicious clients. However, Krum discards the majority of updates and converges at a sub-optimal statistical rate; its quadratic computation complexity also makes it impractical for large participant counts. Multi-objective optimization techniques applied to fuzzy deep learning models offer an alternative to purely distance-based aggregation: by simultaneously optimizing classification accuracy, robustness, and computational efficiency through Pareto-front search, these approaches yield aggregation-layer models that degrade gracefully under partial Byzantine corruption [143]. Coordinate-wise Median and Trimmed Mean [72] achieve near-optimal statistical efficiency under sub-Gaussian noise assumptions while tolerating up to  $f < n/2$  Byzantine clients, but fail against sophisticated model poisoning attacks that can construct gradient updates that lie within the non-trimmed region of the distribution.

Centered Clipping [73] represents a significant theoretical advance, achieving information-theoretically near-optimal Byzantine resilience by iteratively clipping updates that deviate from a reference gradient. In practice, however, it requires a reliable reference gradient—which must itself be computed in a Byzantine-robust manner—and is sensitive to non-IID data distributions that make the reference gradient a poor representative of the honest gradient population. FLTrust [74] addresses this limitation by maintaining a small, trusted clean dataset at the server and computing a trust score for each client update based on its cosine similarity with the server's own gradient update; updates with low trust scores are clipped or excluded. FLTrust has demonstrated empirically that it can maintain model accuracy under up to 90% Byzantine participation—a remarkable result that exceeds the theoretical  $f < n/2$  bound of prior methods—but its security guarantee is contingent on the integrity of the server's clean dataset and its computational isolation from the participants. Interpretability-based extensions to privacy-preserving FL provide a complementary lens for robustness: by making each client's gradient contribution locally explainable, anomalous updates from Byzantine participants become detectable through interpretability divergence rather than purely statistical criteria, an approach validated in industrial cyber-physical system settings [125]. Empirical comparisons of XAI tools across deep learning architectures confirm that no single explainability method generalizes uniformly, making the selection of the appropriate tool a security-relevant design decision when interpretability is used as a Byzantine detection signal [154]. Autonomous intrusion detection systems for vehicular networks demonstrate that self-supervised adaptation of detection thresholds can maintain high detection rates as the vehicle population and threat landscape evolve over time, an architectural principle directly applicable to FL systems in which the participant population is non-stationary [138].

A critical observation that structures the entire Byzantine robustness literature is what Fung et al. [77] term the fundamental limitation of Byzantine-robust aggregation: no aggregation rule can simultaneously satisfy Byzantine tolerance for an arbitrary fraction of malicious clients, convergence to the optimum on non-IID data, and robustness against adaptive adversaries who know the aggregation rule in use. This im-

possibility result implies that the search for a universally effective Byzantine-robust aggregation rule is fundamentally misguided; instead, the field must develop adaptive defenses that respond to observed attack strategies and provide certified robustness guarantees that bound the worst-case model deviation under a specified adversary model. Table IX presents the systematic comparison of Byzantine-robust aggregation methods reviewed in this analysis.

## XI. OPEN CHALLENGES AND RESEARCH DIRECTIONS

The analysis presented in the preceding sections reveals that the field of decentralized and agentic AI security, while considerably advanced relative to five years ago, remains structurally immature. The most significant obstacles to secure real-world deployment are not merely engineering challenges amenable to incremental optimization; they represent fundamental tensions with information-theoretic underpinnings that will require principled, multi-disciplinary research programs to resolve. This

**TABLE IX. BYZANTINE-ROBUST AGGREGATION METHODS: THEORETICAL GUARANTEES AND EMPIRICAL PERFORMANCE**

| Method            | Core Mechanism                                                        | Byzantine Tolerance                   | Statistical Efficiency                      | Key Limitation                                                             |
|-------------------|-----------------------------------------------------------------------|---------------------------------------|---------------------------------------------|----------------------------------------------------------------------------|
| FedAvg (baseline) | Weighted average of client updates                                    | None (vulnerable to single Byzantine) | Optimal for IID data                        | No Byzantine resilience; degrades severely under poisoning                 |
| Krum              | Select update closest to its $n-f$ nearest neighbors by L2 distance   | $f < n/2$ Byzantine clients           | Sub-optimal; discards most updates          | Fails against colluding attackers; high computation for large $n$          |
| Coord. Median     | Coordinate-wise median across client updates                          | $f < n/2$                             | Near-optimal under sub-Gaussian noise       | Fails against sophisticated model poisoning; accuracy loss on non-IID data |
| Trimmed Mean      | Remove extreme fraction $\beta$ from each end; average remainder      | Requires known $f$ ( $\beta = f/n$ )  | Near-optimal                                | Requires known Byzantine fraction; sensitive to $\beta$ tuning             |
| Centered Clipping | Clip updates deviating from reference gradient; iterative refinement  | Informationtheoretically nearoptimal  | Near-optimal                                | Requires reference point; sensitive to non-IID distributions               |
| FLTrust           | Server computes trust score via cosine similarity with clean gradient | Demonstrated: 90% Byzantine           | Good; lower than plain FedAvg on clean data | Requires clean server dataset; potential inference leakage                 |

section characterizes the six most critical open challenges and maps the most promising research directions for each.

The *scalability versus security trade-off* represents the most practically consequential challenge for near-term deployment. Current MPC and HE protocols impose computational overhead that routinely exceeds the utility of the model being protected; deploying BatchCryptstyle FHE-based aggregation, for instance, increases training time by two to three orders of magnitude relative to plaintext FedAvg. The path forward is not solely through hardware acceleration—though domain-specific accelerators such as FPGA-accelerated Microsoft SEAL and Google's Coral TPU are important enablers— but through the development of tight Pareto characterizations that allow system designers to mathematically quantify the security-performance cost of their architectural choices and optimize within acceptable bounds [23], [31], [96], [107]. Decentralized collaborative learning protocols that eliminate model sharing entirely—relying instead on query-based gradient estimation under adversarial heterogeneity—represent a nascent architectural direction that side-steps the communication-overhead bottleneck while maintaining formal convergence guarantees [180].

The *privacy versus utility trilemma* is the most analytically fundamental challenge. The 10–30% accuracy degradation imposed by DP under strong privacy budgets ( $\epsilon \leq 1$ ) is unsustainable for

production deployment in most enterprise applications. The current approach of injecting noise with magnitude proportional to the global sensitivity of the gradient function ignores the internal structure of the neural network—the fact that different layers and parameters contribute differently to privacy risk and to model accuracy. A shift toward gradient-structure-aware DP, in which the noise injection is tailored to the topology and sensitivity distribution of specific layers, offers a promising avenue for substantially reducing the accuracy cost while maintaining formal privacy guarantees [21], [22], [24], [94]. An orthogonal approach that directly addresses the right-to-erasure requirement introduced by GDPR and similar regulations is machine unlearning: Kronecker-factored tensor network approximations enable efficient removal of individual data points’ influence from trained models, making privacy compliance practical without full model retraining [179]. Comprehensive secure and privacy-preserving ML frameworks for industrial IoT environments demonstrate that gradient-structure-aware mechanisms can be instantiated end-to-end within a single deployable system, providing empirical evidence that the trilemma is tractable rather than fundamental when domain constraints are incorporated into the mechanism design [170].

The *benchmarking gap* represents a methodological crisis that undermines the scientific validity of much of the current literature. Attack evaluations in the reviewed literature use incomparable dataset splits, different non-IID distribution parameters, different Byzantine fractions, and different evaluation metrics, making it impossible to determine whether a defense that appears effective in one paper's evaluation would hold under the experimental conditions of another. We echo the imperative need for a community-driven benchmark initiative, analogous to MLPerf for model performance, but specifically focused on adversarial robustness and privacy evaluation in federated learning. Such an initiative should provide open, reproducible attack libraries, standardized non-IID data partitioning protocols, and a common suite of threat models against which defenses are evaluated [6], [77], [108]. A metric-driven gap analysis of current AI security standards reveals that most existing frameworks fail to specify quantitative security properties for distributed ML systems, leaving practitioners without objective criteria for evaluating whether a deployed defense meets the security requirements of their threat model [159].

*Agentic system security* is the most urgent frontier identified in this review, yet it is the area in which the gap between the sophistication of attacks and the maturity of defenses is largest.

We currently lack even the basic formal vocabulary for behavioral monitoring in long-horizon agents: there is no accepted definition of behavioral integrity for an LLM-based agent, no standardized threat model for memory poisoning or indirect prompt injection, and no containment architecture that can limit the blast radius of a compromised agent without fundamentally restricting its autonomous capability. The goal must be to design sandboxing and containment mechanisms analogous to operating system process isolation—mechanisms that allow the agent to operate with full capability within a defined boundary while preventing any compromise from propagating to the broader system or to the federated training process [14], [67], [68], [69], [70]. Structured research trajectories for autonomous cybersecurity identify agent behavioral monitoring and adaptive red-teaming as the two most tractable near-term directions for closing this gap [126]. Red-teaming methodologies based on chain-of-thought reasoning and reinforcement policy optimization have begun to operationalize adversarial evaluation of LLM-based agents, providing the first reproducible protocols for quantifying agent-level vulnerability [127]. Complementary automated red-teaming pipelines that chain reasoning steps with reinforcement-based prompt optimization demonstrate that multi-step attack trajectories can be discovered with substantially fewer human analyst hours than manual red-teaming, pointing toward scalable adversarial evaluation as a production security practice [178].

*Real-time cryptography for AI* represents a challenge whose resolution depends substantially on hardware progress. HE and MPC are too slow for latency-sensitive AI workloads and entirely impractical at LLM scale in their generic forms. Specialized cryptographic hardware—FPGA accelerators, secure enclaves, and application-specific integrated circuits—combined with hybrid architectures that apply full cryptographic protocols only to the most sensitive data-aggregation

pathways while relying on Trusted Execution Environments (TEEs) for lower-sensitivity operations, represent the most promising near-term path [31], [33], [95], [96].

The *Byzantine arms race* reflects the fundamental impossibility result identified by Fung et al. [77]: no aggregation rule can simultaneously satisfy Byzantine tolerance for an arbitrary fraction of malicious clients, convergence optimality on non-IID data, and robustness against adaptive adversaries. This implies that the field must develop adaptive defenses that respond to observed attack strategies and provide certified robustness guarantees bounding the worst-case model deviation under a specified adversary model [71], [72], [73], [74], [107]. A seventh challenge, not captured in Table X, concerns entirely new deployment contexts for agentic AI: immersive extended-reality systems expose agents to sensory-manipulation and spatial-disorientation attack vectors that have no analog in text-domain security, and current standards substantially underspecify the security properties required for such deployments [151]. Table X summarizes all six open challenges.

## XII. DISCUSSION

### A. The Privacy-Security-Utility Trilemma

Our review identifies a pervasive trilemma that haunts the landscape of decentralized intelligence: the fundamental three-way tension between privacy, security, and utility. This is not merely a transient engineering hurdle amenable to incremental optimization; it is a profound information-theoretic trade-off rooted in the basic structure of statistical learning. Privacy mechanisms such as DP, HE, and MPC are designed to mask or aggregate individual signals, yet these very signals are the lifeblood of anomaly detection and Byzantine-resilience: distinguishing a malicious gradient from a non-IID gradient requires exactly the individual-level information that privacy mechanisms are designed to suppress. When security mechanisms are blinded by privacy constraints, the system becomes inherently fragile. This coupling manifests concretely across the mechanisms analyzed in Sections VI and VII. DP-SGD, positioned at the privacy vertex of the trilemma, achieves strong formal guarantees at the cost of 10–30% accuracy degradation—a loss that is unacceptable in production contexts. Secure Aggregation, positioned at the intersection of privacy and utility, provides cryptograph-

**TABLE X. OPEN RESEARCH CHALLENGES: CHARACTERIZATION AND PRIORITY ASSESSMENT**

| Challenge                | Core Issue                                                                                  | Urgency     | Most Promising Directions                                                        | Key References       |
|--------------------------|---------------------------------------------------------------------------------------------|-------------|----------------------------------------------------------------------------------|----------------------|
| Scalability vs. Security | Security mechanisms impose prohibitive overhead at FL training scale                        | High        | Hardware acceleration; approximation; tight Pareto characterization              | [23,31,96,107]       |
| Privacy vs. Utility      | DP noise degrades accuracy; strong $\epsilon$ impractical in non-IID FL                     | High        | Gradient structure exploitation; personalized DP; adaptive clipping              | [21,22,24,94]        |
| Standardized Benchmarks  | Incomparable evaluations; no agreed threat models for agent security                        | Medium-High | Community benchmark initiative; reproducibility standards; open attack libraries | [6,77,108]           |
| Agentic System Security  | Unbounded attack surfaces; memory persistence; cascaded compromise                          | Very High   | Formal agent threat models; session isolation; behavioral monitoring             | [14,67,68,69,70]     |
| Real-Time Cryptography   | HE/MPC too slow for latency-sensitive AI; impractical for LLM scale                         | High        | Specialized cryptographic hardware; hybrid architectures; approximate HE         | [31,33,95,96]        |
| Byzantine Arms Race      | No universally effective aggregation rule; adaptive adversaries always evade known defenses | High        | Adaptive defenses; theoretical lower bounds; certifiable robustness              | [71,72,73,74,77,107] |

|                      |                                                                      |        |                                                                        |
|----------------------|----------------------------------------------------------------------|--------|------------------------------------------------------------------------|
| Regulatory Alignment | Technical mechanisms do not map cleanly to legal privacy definitions | Medium | Multi-disciplinary collaboration; [94] auditable compliance mechanisms |
|----------------------|----------------------------------------------------------------------|--------|------------------------------------------------------------------------|

ic server-blindness without accuracy loss but offers no Byzantine protection whatsoever, leaving the system entirely exposed to model poisoning. FHE, which could theoretically span all three dimensions simultaneously, is rendered impractical by its  $100\times\text{--}1,000\times$  computational overhead. No existing mechanism occupies the interior of the trilemma space where all three properties are simultaneously strong.

Future research must move away from the current sequential optimization approach—in which privacy mechanisms are designed in isolation and then bolted onto an existing security architecture—and toward joint frameworks that treat privacy, security, and utility as a single, multi-dimensional optimization problem with formal trade-off characterizations. A promising direction is differentially private Byzantine-robust aggregation that calibrates noise not to the global sensitivity of the gradient function but to the empirical distribution of honest gradients, exploiting the statistical structure of the nonIID distribution to reduce noise magnitude while maintaining formal privacy bounds. Personalized federated learning [94] offers a complementary approach by decoupling the global model's privacy requirements from those of individual clients, enabling fine-grained noise calibration that is orders of magnitude more efficient than uniform global DP.

### ***B. From Static Models to Dynamic Agents: A Paradigm Shift in Threat Modeling***

The evolution from static, query-response models to action-taking, autonomous agents represents a qualitative rupture in the nature of AI security that existing theoretical frameworks are fundamentally ill-equipped to address. Static model threat surfaces are effectively contained by the boundary of the inference API: an adversary can query the model and observe its outputs, but cannot directly influence its internal state or its environment. In contrast, agentic threat surfaces are theoretically unbounded, scaling with each new tool, API endpoint, or persistent memory bank the agent utilizes.

We find that current theoretical frameworks— Byzantine fault tolerance, basic differential privacy, and standard anomaly detection—were engineered for the static era and do not gracefully extend to the dynamic, goal-directed domain of autonomous agents. The governance dimension of this inadequacy is equally urgent: federated AI systems that serve diverse populations require community-centred design protocols that incorporate affected stakeholders' lived-experience knowledge alongside technical threat modeling, ensuring that equity failures are surfaced before deployment rather than after [160]. Systematic infrastructure inequities—disparities in hardware capability, network connectivity, and data volume across participant institutions—produce models that perform well for resource-rich participants and poorly for under-represented populations, a structural bias that governance frameworks must address as a first-class security concern [161]. The defining characteristic of agentic threats is their temporal diffusion: memory poisoning does not produce an anomalous individual update detectable by Byzantine-robust aggregation; instead, it corrupts the agent's decision-making process over multiple sessions, and its effect on federated gradient updates is spread across many training rounds, making it statistically indistinguishable from legitimate non-IID variation. The cascaded compromise scenario identified in Section V represents the most severe manifestation of this property: a behavioral drift in a single poisoned agent can propagate silently into global model corruption through the gradient communication channel, defeating both agent-level containment and FL-level anomaly detection simultaneously.

Addressing this compound threat requires a joint security architecture that monitors behavioral consistency across agent sessions while simultaneously applying Byzantine-robust gradient filtering at the aggregation layer. At the agent level, the most promising approach involves session-scoped memory isolation—restricting the agent's long-term memory access to read-only during federated training rounds— combined with cryptographic attestation of the agent's decision provenance. At the FL level,

behavioral fingerprinting of gradient updates— tracking the statistical signature of each client's contributions over multiple rounds—can detect the slow drift characteristic of memorypoisoned agents, provided that the fingerprinting mechanism itself is privacy-preserving.

### ***C. The Role of Hardware in Unlocking Cryptographic Deployment***

The computational overhead that currently renders advanced cryptography impractical for production AI deployment is not a fundamental mathematical barrier but a transitory state driven by the immaturity of domain-specific hardware. The emergence of accelerators such as FPGA-accelerated Microsoft SEAL, Intel HEXL, and Google's Coral TPU signals a clear trajectory toward hardware-enabled cryptographic AI. As these platforms mature, the primary bottleneck will shift from raw computational throughput to the trust assumption layer: who controls the hardware, under what security model, and with what verification mechanisms.

The most promising near-term architecture identified in this review is a hybrid strategy that utilizes Trusted Execution Environments (TEEs) for low-overhead, hardware-rooted trust in standard computation, while reserving full cryptographic protocols for the most sensitive dataaggregation pathways where the formal guarantee is non-negotiable. This hybrid approach accepts a stratified security model in exchange for practical deployability. The critical engineering challenge is the secure composition of TEEbased and cryptographic guarantees: an attacker who can compromise the TEE's attestation infrastructure can undermine the security of the entire hybrid system, meaning that the trust assumptions of both components must be carefully analyzed and their failure modes understood before deployment. The co-evolution of cryptographic primitives and AI capabilities is itself a dynamic worth tracking: advances in cryptographic proof systems can enable new classes of verifiable AI computation, while improvements in AI training efficiency can make previously impractical cryptographic training protocols feasible, suggesting a virtuous cycle in which each field accelerates the other [176].

### ***D. Implications for Practice***

Based on the synthesis of 115 studies, we identify four concrete implications for practitioners deploying decentralized AI systems at scale. First, federated learning should never be treated as a standalone privacy solution; it must be combined with at least one formal privacy mechanism (DP or SecAgg) appropriate to the threat model of the specific deployment context. Second, organizations deploying LLM-based agents in any federated context should treat agent memory as a security-critical component requiring integrity protection equivalent to that applied to model parameters, including cryptographic write-authentication for all memory operations. Third, practitioners should adopt PQC standards (KYBER, Dilithium) for new deployments now, regardless of current quantum computing capabilities, given the long operational lifetimes of federated systems and the harvestnow, decrypt-later threat. Fourth, the absence of standardized benchmarks means that reported defense performance figures should be interpreted with significant caution; practitioners should independently reproduce evaluations under conditions matching their specific non-IID distribution before relying on published accuracy numbers. A fifth implication emerges from public health deployments: federated architectures that eliminate the need to share raw population health data across provincial or institutional boundaries unlock collaborative modeling at national scale while preserving data sovereignty, suggesting that regulatory investment in FL infrastructure can yield compounding public-health dividends [128]. Sixth, when LLM-based agents are deployed in health or high-stakes social settings, governance frameworks grounded in equity and accountability principles must be established before, not after, deployment, as the failure modes of under-governed AI agents are asymmetrically borne by vulnerable populations [129].

## **XIII. THREATS TO VALIDITY**

As with all systematic literature reviews, this study is subject to several threats to validity that must be considered when interpreting its findings.

### **A. Internal Validity**

Although seven major databases were searched, gray literature, industry technical reports, and papers published in smaller regional venues may not have been captured. The restriction to English-language publications (12 papers excluded on this basis) may introduce geographic bias by under-representing contributions from non-English-speaking research communities, particularly given the significant volume of FL security research published in Chinese-language venues. The temporal lower bound of 2017 may have excluded foundational work in differential privacy and cryptography published before the FL era that remains highly relevant. The assignment of papers to thematic pillars was performed by two researchers with resolution through consensus ( $\kappa = 0.82$ ); however, the categorization of papers at thematic boundaries inevitably involves subjective judgment.

### **B. External Validity**

The threat models and defensive mechanisms synthesized in this review reflect the state of the literature as of March 2025. The rapid pace of development in LLM-based agentic AI means that new attack classes and defensive frameworks may have emerged since the literature search was completed. Practitioners should treat the research roadmap in Section XI as a directional guide rather than a complete or static characterization of the field. The performance figures cited in Tables VI and VII (accuracy degradation under DP, cryptographic overhead multipliers) are drawn from specific benchmark settings that may not generalize to all deployment scenarios; in particular, the nonIID degree of data heterogeneity varies substantially across real-world deployments and has a strong moderating effect on DP accuracy results.

### **C. Construct Validity**

There is a well-documented tendency in the machine learning literature for positive results to be more frequently published than negative results. This publication bias means that defense mechanisms that fail under certain conditions may be systematically under-represented in the included literature, potentially leading to an overly optimistic picture of the effectiveness of existing defenses. We have attempted to mitigate this by explicitly documenting the conditions under which each defense mechanism fails, as reported in the reviewed papers. Furthermore, the 14 papers in the Agentic AI Security cluster, while representing the complete available literature on this emergent topic, constitute a relatively thin evidential base for strong conclusions; the research recommendations in Section XI should be interpreted accordingly.

## **XIV. CONCLUSION**

This systematic literature review has synthesized findings from 115 peer-reviewed studies published between 2017 and 2025, providing the first comprehensive, PRISMA 2020-compliant analysis of the privacy and security landscape at the intersection of decentralized AI and autonomous agentic systems. Five principal conclusions encapsulate the current state of the field and its most critical future requirements.

First, federated learning, while a substantial advance over centralized data aggregation, is not a privacy panacea. Gradient inversion and membership inference attacks prove conclusively that model updates contain exploitable individual-level information, necessitating rigorous, formal privacy layers beyond the basic FL protocol.

Second, agentic AI has introduced a qualitatively novel threat class—memory poisoning, cascaded tool misuse, and behavioral drift—that has no direct analog in traditional machine learning security. These threats require a dedicated, agent-first security research agenda with new formal threat models, new evaluation methodologies, and new architectural containment mechanisms.

Third, cryptographic primitives (MPC, HE, ZKP) offer mathematically grounded security guarantees but are currently tethered to offline or batch-processing workflows by computational overhead that is prohibitive for interactive AI training. Post-Quantum Cryptography, standardized by NIST in 2024,

must be prioritized in long-horizon system designs to preempt the harvest-now, decrypt-later threat posed by advancing quantum computing capabilities.

Fourth, blockchain-based governance provides unmatched auditability and incentive alignment for federated learning systems, but its current scalability and energy costs preclude deployment in high-frequency training environments. The most viable near-term deployment model restricts the blockchain to low-frequency governance functions while maintaining high-frequency gradient communication off-chain. The broader Web3 ecosystem presents both opportunities and structural challenges for decentralized AI governance: tokenized incentive mechanisms can align participant behavior with system-wide security objectives, but smart contract immaturity, regulatory uncertainty, and market volatility introduce systemic risks that must be explicitly modeled in any production deployment [150].

Fifth, and most fundamentally, the field's primary obstacle is not a lack of individual defensive techniques but a lack of standardized evaluation frameworks that would allow those techniques to be compared, validated, and deployed with confidence. Without common benchmarks, unified threat models, and reproducible evaluation protocols, the arms race between attacks and defenses remains anecdotal rather than scientific. The convergence of decentralization and autonomy represents arguably the most significant security challenge of the next decade. Addressing it will require a multidisciplinary fusion of distributed systems, cryptography, regulatory science, and AI safety, guided by the structured research roadmap delineated in this review.

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